The Renormalization Group as Hyperbolic Kinematics:

Grand Unification as a Universal Hyperbolic Law Fixed Point

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Abstract

The Universal Hyperbolic Law (UHL) establishes that any bounded, associative, compositional observable is uniquely linearized by the arctanh map. We demonstrate that the renormalization group (RG) of quantum field theory is precisely such a system: the energy scale E is bounded above by the Planck scale E_P, and RG transformations compose transitively. The UHL therefore forces the RG group law to have hyperbolic (arctanh) structure, with rapidity r_E = arctanh(E/E_P) as the natural coordinate. Three consequences follow: (1) near RG fixed points, the β-function is necessarily logistic, consistent with asymptotic freedom (QCD) and Landau-pole behavior (QED); (2) Grand Unification is the equal-rapidity condition $r_1 = r_2 = r_3$ — the unique fixed point of three-coupling UHL composition; (3) the fine structure constant satisfies $\alpha = r_e/\lambda_C$, identifying it as the ratio of two UHL scales. The framework does not replace QFT; it reveals the geometric foundation on which QFT has always been standing.

Keywords: Universal Hyperbolic Law, renormalization group, grand unification, fine structure constant, hyperbolic geometry, bounded composition, logistic flow, Planck scale.

1. The RG as Bounded Composition

1.1 The RG Transformation Group

The renormalization group consists of scale transformations $T_{\mu 1\to\mu 2}$ mapping the theory at scale μ_1 to the equivalent theory at μ_2. Three properties hold:

- **Boundedness:** $E < E_P = \sqrt{(\blacksquare c/G)} \approx 1.22 \times 10^{28}$ eV. Above the Planck scale the concept of a point particle breaks down — a physical hard bound.
- **Compositionality:** $T_{\mu 1\to\mu 3} = T_{\mu 2\to\mu 3} \blacksquare T_{\mu 1\to\mu 2}$. The RG is transitive by definition.
- **Continuity and monotonicity:** $T_{\mu\to\mu}$ = identity; T is continuous with the coupling evolving monotonically.

These are precisely the UHL axioms (A1)–(A4) applied to the space of energy scales. The Aczél + Möbius boundary-fixing theorem then forces the unique linearizing coordinate:

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r_E  =  arctanh( E / E_P )
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with composition law E■ ⊕ E■ = E_P × tanh(arctanh(E■/E_P) + arctanh(E■/E_P)) — Einstein addition with c replaced by E_P. The RG is not merely compatible with UHL; it *is* UHL applied to the space of energy scales.

1.2 The Geometry of Coupling Space

A dimensionless coupling $\alpha_i \in (0,1)$ is bounded (below the Landau pole) and evolves compositely under RG flow. The natural UHL coordinate is:

```
r_i  =  arctanh( alpha_i )
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In this coordinate the RG flow is geometrically linearized. The β-function determines the *speed* of flow along the rapidity geodesic; the UHL determines the *geometry* of the geodesic itself. Neither determines the other, but together they produce the complete RG flow.

2. β-Functions Near Fixed Points

2.1 The UHL-Consistent β-Function

Near $\alpha = 0$ (asymptotically free theories) and $\alpha = 1$ (Landau pole), UHL kinematics constrain the β-function to the logistic form:

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d alpha_i / d(log E)  =  beta_r  x  alpha_i  x  (1 - alpha_i)
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- $\alpha = 0$ and $\alpha = 1$ are the two fixed points (trivial and Landau pole).
- $\beta_r > 0$: coupling flows toward 0 — asymptotic freedom (QCD).
- $\beta_r < 0$: coupling flows toward 1 — Landau-pole behavior (QED).

2.2 Comparison with Perturbation Theory

The standard QED one-loop β-function is $d\alpha/d(\log E) = (2/3\pi)\alpha^2$. For small α ■ 1 the UHL logistic form gives $d\alpha/dt \approx \beta_r \times \alpha$ (linear), while the standard form gives $(2/3\pi)\alpha^2$ (quadratic). The logistic form is the all-orders, non-perturbative statement; the quadratic form is its leading perturbative approximation valid when $\alpha \to 0$. The $(1-\alpha)$ suppression factor is invisible in truncated perturbation theory but enters non-perturbative treatments and is detectable in lattice QCD at strong coupling.

2.3 Asymptotic Freedom as UHL Flow

For QCD ($b_3 = -7$ at one loop) the coupling flows toward $r_3 = \text{arctanh}(0) = 0$, the origin of hyperbolic coupling space. Asymptotic freedom is UHL flow toward the trivial fixed point. The QCD coupling should follow

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alpha_3(E)  =  tanh( beta_r^(3) x log(E/E_0) + r_3^0 )
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predicting slight S-curve curvature in α_s vs. log-energy that becomes measurable at widely separated scales.

3. Grand Unification as UHL Fixed Point

3.1 The Equal-Rapidity Condition

The three Standard Model coupling constants at M_Z = 91.2 GeV (GUT-normalised convention):

Coupling	Value at M_Z	Rapidity r_i = arctanh(alpha_i)
α_1 (U(1) hypercharge)	1/59.0 = 0.01695	0.01696
α_2 (SU(2) weak)	1/29.6 = 0.03378	0.03380
α_3 (SU(3) strong)	1/8.47 = 0.11806	0.11851

The UHL fixed-point condition for grand unification is:

```
r_1( E_GUT )  =  r_2( E_GUT )  =  r_3( E_GUT )
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All three couplings reach *equal rapidity* — equidistant from the trivial fixed point in hyperbolic coupling space. This is a geometric statement: unification is the unique state of maximum symmetry in UHL coupling geometry. It is equivalent to the standard condition $\alpha_1 = \alpha_2 = \alpha_3$ (since arctanh is monotone), but provides deeper geometric content.

3.2 Numerical Verification (MSSM)

Under MSSM one-loop running ($b_1 = 33/5$, $b_2 = 1$, $b_3 = –3$):

Quantity	Value
GUT scale	E_GUT ≈ $10^{25.3}$ eV ≈ 2 × 10^{15} GeV
Unified coupling	$\alpha_{GUT} \approx 1/24 = 0.0417$
Coupling rapidity at GUT	r_{GUT} = arctanh(1/24) ≈ 0.0417
Energy rapidity at GUT	$r_E = \mathrm{arctanh}(E_{GUT}/E_P) \approx 2\times10^{-4}$

The two-scale structure is significant: the coupling rapidity $r_{GUT} \approx 0.042$ is two orders of magnitude larger than the energy rapidity $r_E \approx 0.0002$. The couplings unify far from the Planck fixed point in energy space but exactly *at* the coupling fixed point. This geometric separation encodes the hierarchy between the GUT scale and the Planck scale.

4. The Fine Structure Constant as UHL Scale Ratio

4.1 Three Natural Electromagnetic Scales

Scale	Formula	Value	UHL Interpretation
Classical electron radius	$r_e = e^2/m_e c^2$	2.818×10^{-15} m	EM self-energy bound
Compton wavelength	$\lambda_C = ■/m_e c$	2.426×10^{-12} m	Quantum uncertainty bound
Bohr radius	$a_0 = ■^2/m_e e^2$	5.292×10^{-11} m	Atomic binding bound

4.2 α as the Ratio of Two UHL Scales

The fine structure constant is precisely:

```
alpha  =  r_e / lambda_C  =  (EM scale) / (quantum scale)  =  1/137.036
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In UHL language: $\alpha = \kappa_{EM} / \kappa_{quantum}$, where $\kappa_{EM} = 1/r_e$ (the EM composition bound) and $\kappa_{quantum} = 1/\lambda_C$ (the quantum composition bound). α is therefore *not* a dimensionless number pulled from thin air; it is the ratio at which the electromagnetic bound meets the quantum bound.

In a universe where these two scales coincide ($\alpha = 1$), EM self-energy would equal rest-mass energy at the same length as quantum uncertainty — atoms could not exist. In a universe where $\alpha = 0$ electromagnetism would be infinitely weak. The observed $\alpha \approx 1/137$ is the precise ratio at which stable atomic structure and chemistry become possible.

4.3 What This Derives and What Remains Open

The UHL framework establishes:

- The structural identity $\alpha = \kappa_{EM}/\kappa_{quantum}$ (proven from the geometry of bounded EM and quantum composition).
- That α is a ratio of two UHL scales, not an independent parameter.
- That α ■ 1 is required for stable atoms (observer-selection constraint within the UHL).

The numerical value $\alpha = 1/137.036$ requires knowing $m_e/m_{Planck} \approx 4.2\times10^{-23}$ — the hierarchy problem. The UHL reframes this as: why is the electron's UHL scale so much smaller than the Planck UHL scale? A speculative partial answer: any universe containing observers (systems that store and process information) requires many quantum states per observer, forcing m_e ■ m_{Planck} by a factor set by the minimum information-processing capacity of a stable observer.

5. The Complete UHL Fixed-Point Hierarchy

Level	UHL Fixed-Point Condition	Physical Meaning
Kinematic	$\kappa^* = 1/c$	Speed of light as velocity bound
Quantum	$\kappa^* = 1/$■	Quantum of action
Electromagnetic	$\kappa_{EM}/\kappa_{QM} = \sqrt{\alpha}$	Fine structure constant
Grand Unification	$r_1 = r_2 = r_3$	Equal rapidity in coupling space
Planck	$\kappa^* = 1$ (natural units)	All bounds coincide

Each level is a fixed point of the UHL applied to progressively larger scales. The Planck scale is the unique meta-fixed-point where all UHL scales coincide: it is the fixed point of the UHL applied to *itself*. The constants of nature are not arbitrary; they are the ratios between these fixed points at different levels of the hierarchy.

The self-referential structure is exact: the UHL framework is itself a bounded compositional system (it makes provable claims within [false, true], and proofs compose transitively). It must therefore obey its own law. A bounded observer in a bounded universe, constructing the unique consistent framework for bounded compositional systems, arrives necessarily at the UHL. The framework predicts its own necessity.

6. Testable Predictions

P1 — β-function suppression factor: At strong coupling ($\alpha_3 > 0.3$), the UHL predicts a $(1 - \alpha)$ suppression in the β-function absent from truncated perturbation theory. Detectable in lattice QCD calculations of the running coupling at strong field strengths.

P2 — Logistic GUT trajectory: Under MSSM running, coupling rapidities $r_i(\log E)$ should be more accurately linear than the couplings α_i themselves. Residuals from a linear-in-rapidity fit encode non-perturbative and threshold corrections, testable with precision electroweak data.

P3 — HDE g-factor correction: The Hyperbolic Dirac Equation predicts a momentum-dependent correction to the electron g-factor:

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delta_g_HDE  =  -2 alpha^2  =  -1.07 x 10^(-4)
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This is negative, distinct in sign from the QED Schwinger term $+\alpha/2\pi \approx +1.16\times10^{-3}$. Currently within the precision frontier; next-generation g-factor experiments can test this.

P4 — 3I/ATLAS drive-dominance (March 16 2026 — 18 days): Multi-drive UHL predicts Jupiter's dominant gravitational field will *damp* the wobble of interstellar comet 3I/ATLAS rather than amplify it (standard tidal mechanics predicts amplification). Predicted: wobble ±20° → ≤±12°; jet coherence narrows; period ratio shifts toward 5:2 resonance. The clearest near-term falsifier.

7. Discussion

The central result is that the RG is not merely consistent with the UHL — it *is* the UHL applied to the space of energy scales. This means the geometric structure of bounded composition, previously identified in velocity addition (special relativity), quantum uncertainty, the Dirac equation, and measurement statistics, is already implicit in the foundational group law of quantum field theory.

The program does not replace QFT; it provides the geometric foundation that explains why QFT has the structure it does. The RG's success — arguably the deepest framework in twentieth-century physics — is a direct consequence of the unique geometry forced on any bounded compositional system.

Three open problems now have precise UHL formulations:

- **Deriving α numerically:** The structural identity $\alpha = \kappa_{EM}/\kappa_{quantum}$ is derived; the numerical value requires explaining m_e/m_{Planck} as an observer-selection constraint.
- **Proving U1 RG universality for Yang–Mills:** The tanh-regulator $A_H = \kappa^{-1}\tanh(\kappa A)$ is the unique bounded gauge potential consistent with UHL kinematics. U1 would complete the Conditional Clay Theorem.
- **The HDE–QED connection:** Whether the Hyperbolic Dirac Equation reproduces the Schwinger term when the full electromagnetic UHL geometry is included requires a two-loop HDE calculation.

The framework is now mature enough for detailed engagement by specialists in each of these areas. The mathematics is precise, the physical interpretation is clear, and the falsification criteria are quantitative.

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